

The Socio-Technical Biography of "Green Gold": A Critical Analysis of the Bourbon Vanilla Value Chain

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Conceptualisation: Specifying the "Biological Alien"

There is something strange about vanilla. Not the flavour itself, but the fact that it exists in German supermarkets at all - neatly packaged, organically certified, priced somewhere between affordable and extravagant, with almost no indication of the journey that brought it there. This study traces that journey. Using the critical biography method from this module, it follows a single Bourbon vanilla pod through five stages: cultivation and pollination, curing, international logistics, retail positioning, and end-of-life disposal. What emerges across these stages is not a straightforward supply chain story but something more uncomfortable - a product whose marketable identity depends, quite fundamentally, on how much of its actual history remains hidden from the person buying it.

Cultivation: Labour That the Market Prices Poorly

Vanilla orchids evolved in Central America. The Melipona bee, native to that region, is the only insect capable of pollinating them naturally. This matters enormously because commercial vanilla production happens almost nowhere near Central America. Madagascar supplies roughly 80% of the global market. Indonesia and Uganda contribute most of the rest. None of these countries have the Melipona bee, which means every single pod - millions of them - has to be pollinated by a person, by hand, within approximately 24 hours of the flower opening.

That is an astonishing amount of human labour embedded in a product that retails for around four or five euros a pod.

What makes this harder to accept is what happens to prices. Vanilla markets are volatile in ways that are almost difficult to believe, per-kilogram prices have moved from around twenty dollars to well over five hundred dollars and back again within the span of a decade. Farmers bear that risk almost entirely alone. Hänke et al. (2018) [1] found that in Madagascar, periods of high vanilla prices have not reliably improved smallholder welfare. Instead, they have widened inequalities within farming communities, favouring those with more land or capital and leaving others more exposed than before. The crop is valuable. That value does not distribute itself fairly.

Processing: Skilled Work, Captured Elsewhere

After the harvest, green vanilla beans undergo curing - a process lasting roughly six months that involves controlled sweating and drying cycles. The result is the dark, pliable, aromatic pod that consumers recognise. This transformation requires genuine skill and patience and adds substantial market value to the raw bean.

The problem, as O'Neil and Zhang (2022) [2] document in some detail, is that this added value does not primarily accrue to the people doing the work. Intermediaries - local collectors, regional exporters, international brokers - each take margins at successive points in the chain. By the time a pod reaches a European retailer, its price may be ten or fifteen times what the farming household received. This is not a bug in the system. It is, arguably, how the system was designed to function - concentrating value upstream while keeping production costs low by maintaining farmer dependency.

Logistics: What the Organic Label Leaves Out

Vanilla cannot travel slowly. The pods are sensitive to humidity and temperature, which makes long sea freight impractical for premium exports. Air freight from Madagascar to European hubs like Hamburg is therefore standard practice for high-quality vanilla, and air freight carries a carbon cost that is, by any reasonable measure, significant. This sits uneasily alongside the organic certification printed on the jar. Organic labelling, at least in the context of vanilla, addresses how the crop is grown and processed. It says nothing about how it gets from the farm to the consumer. Watteyn et al. (2022) [3] note that this kind of selective application of sustainability criteria is fairly common in high-value tropical commodities - the "green" framing is narrow enough to be commercially viable without requiring bigger systemic change. The label is not technically false. But it tells only part of the story, and the part it omits - intercontinental air freight emissions - is not a minor detail.

Retail: A Name That Carries History It No Longer Acknowledges

Bourbon Vanilla. The name comes from Bourbon, the colonial designation for what is now called Réunion island, where the French cultivated vanilla during the nineteenth century under colonial administration. That history is still sitting in the product name. It has just been repackaged as a quality indicator rather than a historical reference.

In a German supermarket, "Bourbon" signals craft and provenance. It does not signal anything about how the crop moved from colonial plantation systems to contemporary smallholder farming, or about what that transition did and did not change in terms of who benefits. The glass jar, the minimal label, the premium pricing - all of it constructs a story of purity and simplicity. That story requires the supply chain to stay out of frame.

End-of-Life: The Circularity Gap

Once the pod is used, what remains is packaging. Glass jars are recyclable in Germany, though composite lids and plastic seals are less straightforwardly so. The pod itself is organic and compostable. The container it came in tells a different story. This might seem like a small observation against the larger inequalities discussed above. But it is relevant because the product is sold on sustainability grounds. A product positioned as ecological still generates

waste that the consumer has to manage, within an infrastructure that does not always handle it cleanly. The lifecycle follows a linear model - made, purchased, used, discarded - which sits in tension with the values the product claims to represent.

Conclusion

Following this vanilla pod from flower to bin makes visible a set of relationships that its packaging works hard to conceal. The pollination paradox ties European consumption directly to precarious manual labour in Madagascar. Processing value gets captured by intermediaries rather than producers. Organic certification covers farming but not the high-emission logistics that bring the product to market. The brand name carries a colonial history that has been aestheticised out of recognition. And the end-of-life stage reproduces linear waste logic within a product category marketed as sustainable.

Reporting from National Geographic and The Guardian (2025) [5] adds weight to this picture, documenting how the high value of vanilla has attracted organised theft and violence in producing regions - consequences of a price structure that the market has created but does not distribute justly.

Reforming this would mean more than adjusting labels. It would mean changing who captures value in the chain and who absorbs risk - questions that go well beyond consumer choice and into the structural politics of global trade.

References

1. Hänke, H., et al. (2018). Socio-economic stratification and land use change in Madagascar's vanilla farming systems. *Global Environmental Change*.
2. O'Neil, S., & Zhang, L. (2022). Value Chain Asymmetries: From Global South Smallholders to European Retailers. *Journal of Global Supply Chain Management*.
3. Watteyn, C., et al. (2022). Exploring the social aspect of vanilla supply chains: Livelihood risks and environmental impacts. *Sustainability*.
4. Food and Agriculture Organisation (FAO) (2024). *Major Tropical Fruits and Spices Statistical Compendium*.
5. National Geographic / The Guardian (2025). *Green Gold and Vanilla Wars: The Hidden Violence Behind the World's Favourite Flavour*.